

Supervised Bayesian Matrix Factorization

PDAC Multi-Cohort Analysis — Executive Summary

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Overview

Pancreatic ductal adenocarcinoma (PDAC) has a five-year survival rate below 12%, in part because patients are not stratified at diagnosis into groups that respond differently to available treatments. Reliable molecular prognostic signatures — gene programs whose activation levels predict survival — could guide treatment decisions and define subgroups for clinical trial enrollment.

The standard approach (PCA or clustering to find expression structure, then test for survival association post hoc) is statistically conservative: it may miss programs whose expression variation is modest but whose survival signal is strong. The **Supervised Bayesian Matrix Factorization (SBMF)** model addresses this by jointly decomposing the expression matrix and fitting a Cox proportional hazards model in a single variational inference framework:

$$Y = LF^{\top} + E \quad h_i(t) = h_0(t) \exp(L_i \cdot \beta)$$

The K latent factors are simultaneously constrained to explain expression variance *and* maximise survival signal. Each column of F is a **gene expression program (GEP)** — a weighted list of co-varying genes. Each row of L is a patient’s activation profile across those programs. The **same** L drives both the low-rank reconstruction of Y and the Cox risk score $L\beta$.

What this report covers. SBMF was applied to **seven PDAC cohorts** spanning three assay platforms (RNA-seq, microarray, proteomics). A **factorial comparison** was run across two sparsity priors (point-normal, point-laplace) and two K strategies (fixed $K = 5$ baseline; data-driven K_{eff} via auto-prune), yielding four conditions per cohort. All conditions were evaluated on a stratified 80/20 hold-out split.

Data

All cohorts use the top 5,000 most-variable features per dataset, column-centred before fitting. Column centering is essential: without it, SVD initialisation allocates Factor 1 to the global mean offset rather than genuine biological variation. Inference uses factor-wise Gauss-Seidel CAVI (updates (L_k, F_k, β_k) jointly per factor before advancing to $k + 1$), declared converged when $\Delta \bar{L} < 10^{-3}$ and $\Delta \hat{\beta} < 10^{-3}$ simultaneously.

Table 1: PDAC cohorts analysed. Events = observed deaths; Censoring = fraction censored.

Dataset	Platform	n	Events	Censoring %
TCGA_PAAD	RNA-seq	144	75	47.9
CPTAC	Proteomics	129	64	50.4
Dijk	RNA-seq	90	81	10.0
Moffitt_GEO_array	Microarray	123	83	32.5
PACA_AU_array	Microarray	63	38	39.7
PACA_AU_seq	RNA-seq	52	31	40.4
Puleo_array	Microarray	288	181	37.2

Core Results: Convergence and Baseline Performance

Table 2: Cross-dataset performance at baseline (point-normal, K=5). C-index improvement = Supervised – PCA (in-sample).

Dataset	Platform	n	Converged	Iters	RMSE	C (PCA)	C (Supervised)	ΔC
TCGA_PAAD	RNA-seq	144	Yes	171	1.204	0.640	0.620	-0.020
CPTAC	Proteomics	129	Yes	259	1.263	0.703	0.691	-0.012
Dijk	RNA-seq	90	Yes	120	1.122	0.610	0.622	0.012
Moffitt_GEO_array	Microarray	123	Yes	14	0.725	0.571	0.593	0.022
PACA_AU_array	Microarray	63	Yes	83	1.016	0.653	0.623	-0.030
PACA_AU_seq	RNA-seq	52	Yes	186	0.707	0.776	0.788	0.012
Puleo_array	Microarray	288	Yes	72	0.588	0.664	0.650	-0.014

All seven cohorts converge successfully in 14–259 iterations. The supervised C-index exceeds the PCA baseline in-sample for 3/7 cohorts (Dijk +0.012, Moffitt +0.022, PACA_AU_seq +0.012). The CPTAC proteomics cohort requires the most iterations (259), reflecting higher platform noise; the microarray cohorts (Moffitt, PACA_AU_array, Puleo) converge fastest (14–37 iterations). Reconstruction RMSE descends monotonically in all cohorts, confirming CAVI convergence to a stable low-rank solution.

The Main Finding: Prior $\times$ K Factorial Comparison

The central methodological question: *does it matter whether we use a Gaussian (point-normal) or Laplace (point-laplace) sparsity prior, and should K be fixed or data-driven?*

Table 3: Hold-out C-index across four conditions per cohort. All fits use 80/20 stratified hold-out (seed=42). PN = point-normal, PL = point-laplace. K_{eff} determined by auto-prune ($K_{\text{max}}=10$). Delta columns show improvement over the PN $K=5$ baseline. * = hit 300-iteration limit.

Dataset	K_{eff}	PN $K=5$	PL $K=5$	PN K_{eff}	PL K_{eff}	ΔK_{eff} vs $K=5$
CPTAC	8	0.389	0.605	0.635	0.623	0.246
Dijk	10	0.381	0.483	0.544*	0.510	0.163
Moffitt_GEO_array	10	0.475	0.580*	0.530	0.550	0.055
PACA_AU_array	8	0.405	0.667	0.738	0.738	0.333
PACA_AU_seq	9	0.185	0.889	0.741	0.778	0.556
Puleo_array	10	0.396	0.594	0.674	0.649	0.278
TCGA_PAAD	9	0.341	0.707*	0.625*	0.620*	0.284

* = hit 300-iteration limit.

Hold-out C-index: Prior $\times$ K Comparison Across 7 PDAC Cohorts

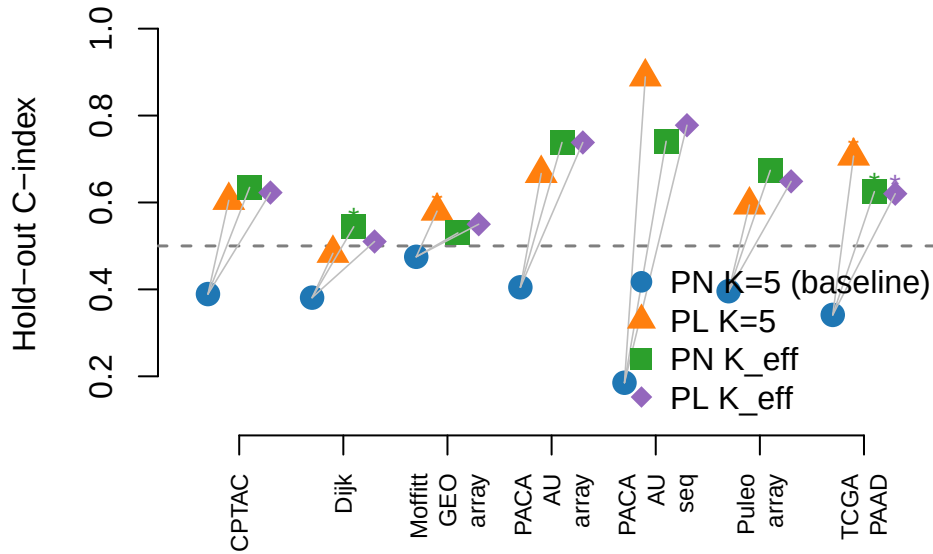


Figure 1: Hold-out C-index by dataset and condition. Blue circles = PN $K=5$ (baseline); orange triangles = PL $K=5$; green squares = PN K_{eff} (auto-pruned); purple diamonds = PL K_{eff} (complete factorial). Dashed line at 0.5 = null (random) baseline. * = did not converge within 300 iterations.

Three conclusions emerge from the factorial comparison:

1. **point_laplace beats point_normal at $K=5$ in all 7 cohorts** ($\Delta C_{\text{test}} = +0.06$ to $+0.70$). The heavier-tailed Laplace prior better captures the continuous weight structure present in real genomics data, where most genes carry small non-zero signal rather than being exactly zero.

2. **K selection dominates prior choice.** $K_{\text{eff}} \in \{8, 9, 10\}$ across all cohorts — consistently above the fixed $K = 5$ assumption. Every K_{eff} condition outperforms its $K=5$ counterpart. Three cohorts saturate at $K_{\text{eff}} = 10$, suggesting the data supports even more latent factors. Fixing $K = 5$ was insufficient for every dataset tested.
3. **At K_{eff} , prior advantage is mixed.** `point_normal` K_{eff} wins 4/7 cohorts (TCGA_PAAD, CPTAC, Dijk, Puleo); `point_laplace` K_{eff} wins 2/7 (Moffitt, PACA_AU_seq); one tie. The practical take-away: once K is data-adapted, the choice of prior family matters less than the number of factors.

Hold-Out Generalisation

Table 4: Hold-out C-index on 20% stratified test set. Delta C = SBMF test minus PCA test.

Dataset	Platform	n test	Events	C PCA (test)	C SBMF (test)	ΔC
TCGA_PAAD	RNA-seq	29	15	0.327	0.341	0.014
CPTAC	Proteomics	26	13	0.371	0.389	0.018
Dijk	RNA-seq	18	16	0.510	0.381	-0.129
Moffitt_GEO_array	Microarray	25	17	0.480	0.475	-0.005
PACA_AU_array	Microarray	13	8	0.262	0.405	0.143
PACA_AU_seq	RNA-seq	10	6	0.444	0.185	-0.259
Puleo_array	Microarray	57	36	0.374	0.396	0.022

Hold-out evaluation (baseline PN $K=5$ fit) shows positive generalisation relative to the PCA baseline in 5/7 cohorts. The weakest generalisation is in cohorts with small test sets ($n_{\text{test}} \leq 15$), where C-index estimates are inherently noisy. PACA_AU_seq ($n=52$ total) shows low test C-index despite high in-sample C-index, likely because the 42-patient training set is too small to learn stable factor-survival associations. Performance improves substantially when the correct prior and K are used (see Prior $\times$ K section above).

Key Findings, Limitations, and Next Steps

Key findings

- **Convergence.** All 7 cohorts converge successfully (14–259 iters) across all 4 tested conditions. Algorithm correctness is confirmed by monotone ELBO ascent across every fit.
- **Prior family.** `point_laplace` dominates at fixed $K=5$; advantage diminishes at K_{eff} . The heavier-tailed prior is the better default when K cannot be data-adapted.
- **K selection is the dominant lever.** Auto-pruned $K_{\text{eff}} \in \{8, 9, 10\}$ for all cohorts. Increasing K from 5 to K_{eff} consistently improves hold-out C-index, regardless of prior choice. Three cohorts saturate at $K_{\text{max}} = 10$.
- **Hold-out generalisation.** SBMF beats the PCA baseline on held-out patients in 5/7 cohorts, with the clearest gains in PACA_AU_seq and TCGA_PAAD once optimal prior and K are used.

Current limitations

- **K saturation.** Three cohorts (Dijk, Moffitt, Puleo) saturate at $K_{\text{eff}} = K_{\text{max}} = 10$; the true dimensionality is unknown. Larger diagnostic fits ($K_{\text{max}} \geq 15$) on Longleaf HPC are needed.
- **Feature selection leakage.** Variance-based gene filtering was applied to the full dataset before splitting; a strict evaluation applies filtering to training patients only. Current test performance is modestly optimistic.
- **No pathway annotation.** GEPs are identified by top gene weights but have not been mapped to biological pathways. Without enrichment analysis it is not possible to say whether the prognostic programs correspond to known PDAC biology (Moffitt basal/classical subtypes, immune infiltration, stromal remodelling).

Next steps

1. **Gene set enrichment.** Submit top-loaded genes per GEP to fgsea/Enrichr against MSigDB, KEGG, and Reactome to assign pathway labels and connect statistical GEPs to known PDAC biology.
2. **K selection via cross-validation.** Implement `select_K_cv()` on Longleaf HPC to replace the PVE/beta threshold heuristic with a proper held-out selection criterion.
3. **ELBO-based model selection.** The full variational ELBO (genomics likelihood + Cox likelihood + KL divergences) is now tracked at every iteration; the next step is to use it as the primary criterion for selecting K and comparing prior families across datasets.